Simulate before you sample: a three-step loop for designing ecological studies

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Abstract

1. Much of what an ecological study can reveal is fixed before any data are collected. Sample size, treatment levels, the range of predictor values and measurement effort together determine which effects can be estimated, and how precisely. Yet these choices are often made by convention, or planned only against a significance threshold, which promotes dichotomous "significant / non-significant" thinking from the outset.

2. Here we propose a practical addition to the research workflow: a *simulation loop*, carried out entirely before data collection. The loop has three steps: (1) translate a conceptual hypothesis into an explicit generative model; (2) simulate and visualise the data that model implies, to scrutinise assumptions about effect sizes, variability and measurement limitations; and (3) iterate design choices—sample size, treatment levels, predictor range, measurement effort—using criteria focused on estimation quality, especially interval width (precision) and bias, rather than binary detect/not-detect framing. We illustrate the loop with worked linear and non-linear examples and reproducible R code, requiring no specialised statistical machinery beyond base R.

3. The examples show how the loop exposes identifiability limits that no increase in sample size can resolve, how precision can depend as much on *where* you sample as on how much you sample, and how design targets can be set from the precision a scientific or management claim actually requires.

4. Simulating before sampling does not guarantee correct inference; its value is conditional on the assumptions explored. But making those assumptions explicit and inspectable becomes increasingly important as ecological analyses draw on larger datasets and AI-assisted pattern discovery, where deciding what to believe depends on clearly stated assumptions and defensible uncertainty quantification. We intend the loop as a teachable entry point for researchers and students who have never written a generative model.

Keywords: statistical methods, quantitative ecology, hypothesis testing, scientific reasoning, experimental design, power analysis, study design, simulation, statistical power, precision, effect size, reproducibility.

1 Introduction

Many of us have seen the same scene in a conference talk, an in-house seminar, or a paper. A speaker presents a compelling story, shows a clean figure with the mean response in a control and a treatment group, and points to the familiar stars of significance, concluding that the treatment "has an effect" ($p < 0.05$)—or, more problematically, that it "has no effect" ($p > 0.05$). The combination of a coherent narrative, an intuitive plot and a thresholded summary can make the conclusion feel settled, even though the same data are often compatible with a wide range of effect sizes. For many listeners and readers, this leaves a quieter set of questions: how much do we actually know from these data, what else are they compatible with, and would the interpretation change under a different but still reasonable analysis?

These concerns are not new. The binary, threshold-based style of statistical reasoning that many of us still use—and are often still taught to use—originated about a century ago (Neyman & Pearson, 1933; Gigerenzer, 2004) and has been criticised for much of that time, with especially clear calls in recent decades to move beyond "statistical significance" as a scientific standard (e.g. Anderson *et al.*, 2000; Amrhein *et al.*, 2017; Wasserstein *et al.*, 2019; Amrhein *et al.*, 2019). Yet the reform has been uneven in ecology and evolutionary biology. Surveys of the published literature find widespread under-powered designs, selective reporting and systematically exaggerated effect sizes (Kimmel *et al.*, 2023; Yang *et al.*, 2023; Fraser *et al.*, 2018), and quantitative training often still centres on the very tests being criticised (Touchon & McCoy, 2016).

Most of this reform effort has been directed at how results are analysed and reported. We think an equally important part of the problem sits earlier. By the time data exist, much of what a study can reveal has already been fixed by choices made months before: how many replicates, which treatment levels, how wide a gradient of conditions to span, and how carefully to measure. No subsequent modelling choice can recover information that the sampling design never collected. Worse, a design with poor precision does not simply fail to detect an effect. Conditional on passing a significance filter, such a design reports effects that are systematically too large, and sometimes of the wrong sign (Gelman & Carlin, 2014). This is a plausible mechanism behind the exaggeration documented in the ecological literature (Kimmel *et al.*, 2023), and it is a property of the design rather than of the analyst.

The conventional tool for making these choices—power analysis—inherits the framing we are trying to escape. It asks whether a test will cross a threshold rather than how precisely a quantity will be known, and it is frequently misapplied after the data are in (Hoenig & Heisey, 2001). Planning instead for *precision*, that is, for an uncertainty interval narrow enough to support the claim at stake, has been advocated for decades (Goodman & Berlin, 1994; Cumming, 2013; Rothman & Greenland, 2018; Maxwell *et al.*, 2008; Lakens, 2022). In ecology, however, precision-based planning meets a practical obstacle: closed-form expressions exist only for the simplest designs, while the settings ecologists actually work in—saturating responses, nested sampling, non-Gaussian data—have none. What is needed is a way to reach the same answers by simulation.

Here we propose a small but practical addition to the research workflow: a *simulation loop*, carried out before any data are collected. The loop has three steps. First, translate a conceptual hypothesis into an explicit generative model, that is, a formula stating how the data would arise, including both the expected response and its stochastic component. Second, simulate and visualise the data that model implies, and ask whether they look like data the system could plausibly produce. Third, iterate the design—sample size, treatment levels, the range of predictor values, measurement effort—judging each version by the precision it delivers for the parameter that matters, rather than by whether a test would clear a threshold. The procedure is a loop because each step can send us back to the previous one: simulated data that look absurd send us back to the model, and a design that cannot deliver useful precision sends us back to the question.

Simulation itself is not new in ecology and evolution, and our contribution is neither the idea nor the machinery but a deliberately narrow and teachable version of them. DiRenzo *et al.* (2023) show how simulation can validate complex models once they have been specified, and Wolkovich *et al.* (2026) set out a four-step workflow spanning model development, fitting to empirical data and post-fit checking, framed in a Bayesian setting; both belong to a broader move towards explicit statistical workflows (Gelman *et al.*, 2020; Schad *et al.*, 2021), and simulation has long been standard for evaluating statistical methods in general (**?**). The loop we describe sits earlier and asks less: it stops before data exist, it takes estimation precision rather than model calibration as its criterion, and it requires no machinery beyond base R. We intend it as an entry point for researchers and students who have never written a generative model, and as a step towards those fuller workflows rather than a substitute for them. Throughout, we provide worked examples with reproducible code, covering a linear case, a non-linear case, a structurally misspecified model, a confounded observational design and a nested design. Figure **??** places the loop within the conventional research workflow and marks the pitfalls it can help anticipate; Box 1 defines those pitfalls and gives key references.

2 The simulation loop

Here we propose a small but practical addition to the traditional scientific workflow (Figure 1): a simulation loop that is implemented before data collection (or early in analysis). The aim is to make assumptions explicit, to translate verbal hypotheses into quantitative expectations, and to explore what the planned data and model can support under realistic variability and measurement limitations.

2.1 Core idea: simulate from an explicit data-generating model

At the centre of the simulation loop is an explicit data-generating model. A useful starting point is to separate the systematic component (pattern) from the stochastic component (noise):

$$y_i = f(x_i; \theta) + \varepsilon_i \tag{1}$$

where $f(x_i; \theta)$ is the expected value of y_i given predictors x_i and parameters θ (the signal implied by the hypothesis), and ε_i is the remaining deviation around that expectation (variation not captured by the model). Depending on context, ε_i may represent process variation, observation/measurement error, or both. Importantly, 'noise' is not inherently meaningless; it is whatever remains after accounting for the structure we chose to model.

In addition to specifying how y is generated conditional on x, it is often useful to specify how x arises under the planned design (e.g. treatment allocation, predictor range, spatial layout, measurement schedule). Making both parts explicit clarifies which features of the data are determined by design choices and which are left to chance. By simulating from the full data-generating process, we can explore what patterns we should expect to see if our hypothesis and assumptions were approximately correct—and what range of outcomes could plausibly arise from variability and measurement limitations alone.

2.2 The three steps of the simulation loop

Step 1: From a verbal hypothesis to a mechanistic (mathematical) model.
Models are hypotheses. With each model we apply to our data, we examine—often without full awareness—an additional hypothesis. To constrain and justify this process to a reasonable set of hypotheses, we propose to build mechanistic models by incorporating existing knowledge about the influencing variables and their relationships in question. Which explanatory variables are influencing y? Is the relationship linear or non-linear? What do we know from the literature? This step nudges us to think carefully about our hypothesis in terms of model parameters, translating subjective and conceptual hypotheses into explicit mathematical formulations. In our experience, a conceptual hypothesis often intuitively makes sense until we try to translate it into a mechanistic model; then it can become

clear that the original hypothesis lacks direction, concreteness, or foundation in basic knowledge. Beyond the classical descriptive hypothesis, a mechanistic model also forces us to think about what we actually want to estimate, and what values we expect for those parameters (e.g. effect sizes and their variance). This gives us the opportunity to state hypotheses with numbers and formulas. In ecology, this often includes anticipating common design features such as hierarchy and non-independence (e.g. repeated measurements, plots nested within sites), and representing them explicitly in the model.

Step 2: Simulate and visualise synthetic ('fake') data.
"The greatest value of a picture is when it forces us to notice what we never expected to see." This quote from Tukey (1977) captures the motivation for the second step. With the mechanistic model sorted out, we generate a synthetic dataset from it. The key point is not to simulate only a marginal distribution for y, but to simulate y through the model structure: we generate predictor values under the planned design, compute the expected response via $f(x;\theta)$, and then add stochasticity via ε to represent variability and measurement limitations. By plotting the generated data, we see our hypothesis in a concrete, visualised form—not only the direction of an effect, but also its magnitude (e.g. effect size) and consistency (variance). We believe that many ecological papers would benefit from visualising mechanistic model predictions to validate and communicate hypotheses more effectively.

Step 3: Model exploration and design iteration.
The third step is to explore what the model and planned design typically produce when we repeat the study many times on the computer. This goes beyond a classical power analysis that targets a significance threshold. Instead, we repeatedly simulate data under plausible assumptions, fit the planned statistical model, and examine parameter estimates and uncertainty. Does the model behave as we think it does? Do the simulated data show realistic variation? How do estimates and uncertainty intervals change with sample size, predictor range, measurement error, or alternative model structures? We can also fit the model to the simulated data and check whether parameter estimates recover the values we used to generate the data. Going back and forth between assumptions, simulation, and fitting is a playful but powerful way to design a study, understand limitations, and avoid overconfident interpretation before analysing empirical data.

In the following sections we provide concrete examples and reproducible R code to illustrate how this procedure can be applied in practice.

3 Practical example (linear): Plant growth and nitrogen

Data simulation is best shown in practice. The example presented here is deliberately simple and can be reproduced directly. A fully commented R script containing all code shown below (including the simulation loops and figures) can be downloaded from the Supplement.

What influences y in what way?

Every study starts with a research question. Here: What is the influence of nitrogen fertilization on plant growth? We might perform an experiment on sunflowers and fertilize some with nitrogen to assess biomass after one season. Our response variable y is biomass, and a verbal hypothesis could be that plants grow more when nitrogen increases.
To translate this hypothesis into a mechanistic (mathematical) model, we assume a positive linear relationship between biomass and nitrogen concentration x. Here, the "mechanistic" component is intentionally simple and phenomenological; later examples extend the same workflow to explicitly non-linear response functions. A simple model is:

$$y_i = a + bx_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma) \tag{2}$$

where a is the intercept (expected biomass at $x = 0$ on the chosen scale), b is the slope (expected biomass increase per unit nitrogen; an effect size), and σ captures variability around the mean relationship (process variation and/or measurement error).

Setting model parameters

This is where things get interesting and concrete: What do we think these model parameters should look like? In other words, what biomass do we expect at low nitrogen, how strong is the nitrogen effect, and how variable is the relationship?
For illustration, we choose:

```
# Specify model parameters
a <- 30     # intercept
b <- 7      # slope (effect size)
sigma <- 5  # SD of residual variation
```

These values are not "truth"; they make our assumptions explicit. Later, we can explore how conclusions change when we vary them.

Simulate!

To simulate, we specify a sample size and generate a plausible range of nitrogen concentrations. Here we treat nitrogen as a continuous gradient:

```
# Choose a design and simulate data
n <- 50
x <- rnorm(n, mean = 10, sd = 4)          # nitrogen concentrations
y <- a + b*x + rnorm(n, mean = 0, sd = sigma)

fake <- data.frame(x = x, y = y)
```

In experiments with fixed treatment levels (rather than a gradient), x would be set to those levels with replication per level, rather than drawn from a distribution. The same simulation logic applies; only the design for generating x changes.

Visualize and assess your fake data

Now we plot the generated data and check whether it matches what we had in mind. This is the hypothesis in visual form, including both effect size and variability.

```
plot(fake$x, fake$y,
xlab = "Nitrogen concentration (x)",
ylab = "Biomass (y)",
main = "Simulated (fake) data")
```

We can also fit the planned model and check whether the estimated parameters are in the range we assumed. This is a quick sanity check that the analysis can recover the generating values under these assumptions.

```
fit_lm <- lm(y ~ x, data = fake)
summary(fit_lm)
confint(fit_lm)  # 95% confidence intervals

plot(fake$x, fake$y,
xlab = "Nitrogen concentration (x)",
ylab = "Biomass (y)",
main = "Simulated data with fitted regression line")
```

```
        abline(fit_lm)
```

At this point, it is tempting to focus on a p-value. In the simulation loop, we instead look at the estimates and their uncertainty—for example, the width of the 95% confidence interval for b.

How to use this — Play!

Now we go back and forth and explore what changes when we manipulate assumptions and design choices. A simple and practical target is precision: how wide are uncertainty intervals for the parameter we care about (here the nitrogen effect b), and how does that depend on sample size, the range of nitrogen values, or the amount of variability and measurement error?
The code below repeats the simulated study many times and summarises the width of the 95% confidence interval for b across repetitions. These design–precision curves are conditional on the assumed model (linearity, error distribution, and constant variance); later examples show how to relax these assumptions and test sensitivity.

```
        # One simulation run: generate data, fit model, return CI width for slope
            b
        one_run <- function(n, a, b, sigma, mean_x = 10, sd_x = 4) {
                x <- rnorm(n, mean = mean_x, sd = sd_x)
                y <- a + b*x + rnorm(n, 0, sigma)
                fit <- lm(y ~ x)
                ci <- confint(fit)["x", ]
                return(ci[2] - ci[1])
        }

        set.seed(1)

        # Precision vs sample size
        ns <- c(10, 20, 30, 50, 80, 120)
        R <- 500

        out_n <- data.frame(n = ns, mean_width = NA, sd_width = NA)
        for (j in seq_along(ns)) {
                widths <- replicate(R, one_run(ns[j], a, b, sigma))
                out_n$mean_width[j] <- mean(widths)
                out_n$sd_width[j] <- sd(widths)
        }

        plot(out_n$n, out_n$mean_width, type = "b",
        xlab = "Sample size (n)",
        ylab = "Mean 95% CI width for slope (b)",
        main = "Precision of the nitrogen effect vs sample size")
```

If a study requires the nitrogen effect to be estimated within $\pm k$ biomass units per nitrogen unit, then a rough design target is a 95% confidence interval width of about $2k$ for b. The simulation loop provides a direct way to choose n (and other design choices) to meet such a target under stated assumptions.
A second lever in regression is the range of the predictor: even with the same n, a narrow nitrogen range makes the slope harder to estimate. We can explore that directly:

```
        # Precision vs predictor range (sd of x), at fixed n
        sd_x_values <- c(1, 2, 4, 6)
        n_fixed <- 50

        out_x <- data.frame(sd_x = sd_x_values, mean_width = NA)
        for (k in seq_along(sd_x_values)) {
```

```
            widths <- replicate(R, one_run(n_fixed, a, b, sigma,
            mean_x = 10, sd_x = sd_x_values[k]))
            out_x$mean_width[k] <- mean(widths)
    }

    plot(out_x$sd_x, out_x$mean_width, type = "b",
    xlab = "SD of nitrogen values (predictor range)",
    ylab = "Mean 95% CI width for slope (b)",
    main = "Precision depends on predictor range")
```

Finally, we can explore how increased variability and measurement error affect precision. For example, if the residual standard deviation doubles, uncertainty intervals widen substantially under the same design:

```
    # Sensitivity to variability / measurement error: larger sigma
    sigma2 <- 10

    out_n2 <- data.frame(n = ns, mean_width = NA)
    for (j in seq_along(ns)) {
            widths <- replicate(R, one_run(ns[j], a, b, sigma2))
            out_n2$mean_width[j] <- mean(widths)
    }

    plot(out_n$n, out_n$mean_width, type = "b",
    xlab = "Sample size (n)",
    ylab = "Mean 95% CI width for slope (b)",
    main = "Precision vs sample size under different variability")
    lines(out_n2$n, out_n2$mean_width, type = "b", lty = 2)
    legend("topright", legend = c(paste0("sigma = ", sigma),
    paste0("sigma = ", sigma2)),
    lty = c(1, 2), bty = "n")
```

This back-and-forth is the core of the simulation loop in practice: we make assumptions explicit, generate the data those assumptions imply, and then explore how design choices translate into the uncertainty of our key estimates. Improving precision, however, does not by itself make an association causal; the loop complements, but does not replace, careful causal reasoning and design. In the next section, we extend the same workflow to non-linear relationships, where model structure and identifiability become even more central.

4 Practical example (non-linear): Saturating growth response to nitrogen

Many ecological relationships are non-linear: responses can saturate, show thresholds, or peak at intermediate conditions. In such cases, simulation is especially useful because the same data can often be explained by different parameter combinations, and identifiability can depend strongly on the predictor range and the design. Below we extend the same workflow to a simple saturating response of biomass to nitrogen, using a Michaelis–Menten (rectangular hyperbola) function.

What influences y in what way?

We keep the same research question—how nitrogen influences plant growth—but now assume that growth increases with nitrogen and then levels off as other resources become limiting. A simple non-linear model is:

$$y_i = \frac{V_{\max} x_i}{K + x_i} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma) \tag{3}$$

where $V_{\max}$ is the asymptotic biomass (maximum expected response as $x \to \infty$), K is the half-saturation constant (the nitrogen value at which expected biomass is $V_{\max}/2$), and σ captures variability and measurement error around the mean response.

Setting model parameters

As before, we must state plausible parameter values. For illustration, we choose:

```
# Non-linear model parameters (Michaelis-Menten)
Vmax <- 120   # asymptotic biomass
K <- 8        # half-saturation constant
sigma <- 10   # residual SD
```

Simulate!

A key design choice in non-linear models is the range of x. To show why this matters, we simulate two designs with the same sample size: one that samples nitrogen only in a low-to-moderate range (where saturation is not well observed), and one that samples a wider range that includes values high enough to approach the asymptote.

```
# Helper: Michaelis-Menten mean function
mu_mm <- function(x, Vmax, K) Vmax * x / (K + x)

set.seed(2)
n <- 50

# Design A: narrow nitrogen range (poor leverage on asymptote)
x_narrow <- runif(n, min = 0, max = 12)

# Design B: wide nitrogen range (better leverage on asymptote)
x_wide <- runif(n, min = 0, max = 40)

# Simulate data
y_narrow <- mu_mm(x_narrow, Vmax, K) + rnorm(n, 0, sigma)
y_wide   <- mu_mm(x_wide,   Vmax, K) + rnorm(n, 0, sigma)

fake_narrow <- data.frame(x = x_narrow, y = y_narrow)
fake_wide   <- data.frame(x = x_wide,   y = y_wide)
```

Visualize and assess your fake data

Plotting the two simulated datasets immediately shows why the design matters: in the narrow range, the relationship can look almost linear, whereas the wide range reveals curvature and the onset of saturation.

```
plot(fake_narrow$x, fake_narrow$y,
xlab = "Nitrogen (x)", ylab = "Biomass (y)",
main = "Non-linear model: narrow nitrogen range")
curve(mu_mm(x, Vmax, K), add = TRUE)

plot(fake_wide$x, fake_wide$y,
xlab = "Nitrogen (x)", ylab = "Biomass (y)",
main = "Non-linear model: wide nitrogen range")
curve(mu_mm(x, Vmax, K), add = TRUE)
```

How to use this — Play!

We now fit the planned non-linear model and examine what it typically produces under each design. In base R, a simple option is **nls** (non-linear least squares). Non-linear fitting can be sensitive to starting values, so we provide reasonable starts.

```
# Fit non-linear model with nls
fit_mm <- function(dat) {
        nls(y ~ Vmax * x / (K + x),
        data = dat,
        start = list(Vmax = max(dat$y), K = median(dat$x)),
        control = nls.control(maxiter = 200, warnOnly = TRUE))
}

fitA <- fit_mm(fake_narrow)
fitB <- fit_mm(fake_wide)

summary(fitA)
summary(fitB)
```

Even when **nls** converges, the key question is not whether parameters exist, but how uncertain they are and how much they depend on design choices. A simple, comparable metric is again interval width. For **nls**, we can approximate confidence intervals for parameters. (In some datasets, `confint` may fail or be slow; in that case, repeating the simulation and examining the spread of estimates provides the same message.)

```
# Approximate 95% confidence intervals for parameters (may be slow for
    some fits)
confint(fitA)
confint(fitB)
```

To make the comparison more robust and aligned with the simulation loop, we repeat the whole study many times under each design and summarise the distribution of parameter estimates. This makes identifiability issues visible: if many different (Vmax, K) combinations fit the data similarly, the estimates will vary widely across repetitions, especially under the narrow-range design.

```
one_run_mm <- function(n, x_max, Vmax, K, sigma) {
        x <- runif(n, 0, x_max)
        y <- mu_mm(x, Vmax, K) + rnorm(n, 0, sigma)
        dat <- data.frame(x = x, y = y)

        fit <- try(fit_mm(dat), silent = TRUE)
        if (inherits(fit, "try-error")) return(c(Vmax_hat = NA, K_hat = NA
            ))

        co <- coef(fit)
        c(Vmax_hat = unname(co["Vmax"]), K_hat = unname(co["K"]))
}

set.seed(3)
R <- 300
n <- 50

# Design A: x_max = 12 (narrow); Design B: x_max = 40 (wide)
estA <- replicate(R, one_run_mm(n, x_max = 12, Vmax, K, sigma))
estB <- replicate(R, one_run_mm(n, x_max = 40, Vmax, K, sigma))

```

```r
estA <- t(estA); estB <- t(estB)
estA <- as.data.frame(estA); estB <- as.data.frame(estB)

# Remove failed fits
estA <- estA[complete.cases(estA), ]
estB <- estB[complete.cases(estB), ]

# Compare distributions of estimates
summary(estA)
summary(estB)

plot(estA$K_hat, estA$Vmax_hat,
xlab = "K estimate", ylab = "Vmax estimate",
main = "Narrow range: strong parameter trade-off")
abline(v = K, h = Vmax, lty = 2)

plot(estB$K_hat, estB$Vmax_hat,
xlab = "K estimate", ylab = "Vmax estimate",
main = "Wide range: improved identifiability")
abline(v = K, h = Vmax, lty = 2)
```

This "cloud of estimates" is a simple diagnostic: in the narrow-range design, $V_{\max}$ and K often trade off against each other (many combinations produce similar curves over $x \in [0, 12]$), so parameter uncertainty can remain large even with the same sample size. In contrast, including higher nitrogen values provides leverage to distinguish curvature and the asymptote, improving identifiability and reducing uncertainty.

Finally, we can connect this directly back to design planning by comparing a precision proxy (e.g. the interquartile range of estimates or the spread of fitted parameters) across choices of n and x-range. For example:

```r
# Quick design exploration: how does n affect the spread of K estimates?
ns <- c(20, 50, 100)
R <- 200

spread <- function(n, x_max) {
        est <- replicate(R, one_run_mm(n, x_max, Vmax, K, sigma))
        est <- t(est)
        est <- est[complete.cases(est), , drop = FALSE]
        # use IQR as a robust spread measure
        c(IQR_Vmax = IQR(est[, "Vmax_hat"]), IQR_K = IQR(est[, "K_hat"]))
}

out <- expand.grid(n = ns, x_max = c(12, 40))
out$IQR_Vmax <- NA
out$IQR_K <- NA

for (i in seq_len(nrow(out))) {
        s <- spread(out$n[i], out$x_max[i])
        out$IQR_Vmax[i] <- s["IQR_Vmax"]
        out$IQR_K[i] <- s["IQR_K"]
}

out

plot(out$n[out$x_max == 12], out$IQR_K[out$x_max == 12], type = "b",
xlab = "Sample size (n)", ylab = "IQR of K estimates",
```

```
        main = "Precision depends on n and x-range")
        lines(out$n[out$x_max == 40], out$IQR_K[out$x_max == 40], type = "b", lty
            = 2)
        legend("topright", legend = c("x in [0,12]", "x in [0,40]"),
        lty = c(1,2), bty = "n")
```

This non-linear example illustrates why simulation is especially valuable beyond linear settings: identifiability and precision can depend as much on *where* you sample (predictor range and design) as on how many observations you collect. In the next section, we generalise these ideas and briefly discuss how simulation-based thinking becomes even more important when models grow more complex (e.g. multiple predictors, interactions, hierarchical designs) and when inference relies on additional modelling assumptions.

5 Practical guidance: implementing the simulation loop

This section summarises practical guidance to implement the simulation loop in everyday ecological research. The goal is not to add another layer of complexity, but to make assumptions explicit, align study design with the parameters we care about, and report simulation results transparently and reproducibly.

5.1 Practical guidance

Choose a generative model that matches the question. Start from the verbal hypothesis and translate it into a simple generative model: define y, the predictors/design variables x, the expected response $f(x;\theta)$, and the stochastic component ε (process variation and/or measurement error). Begin simple and add complexity only when it changes the inference you want to make (e.g. hierarchy, autocorrelation, heterogeneity, detection).
State plausible parameter values (and ranges). Simulation is only as informative as the assumptions it explores. Use prior knowledge, the literature, pilot data, or clearly stated expert judgement to propose plausible values for effect sizes, variance components, and measurement limitations. When unsure, explore a range of values and report how conclusions depend on them.
Simulate from the design you actually plan to run. Explicitly represent the sampling design (treatment levels, replication structure, predictor range, blocking, repeated measures). In non-linear settings, pay special attention to the predictor range: identifiability can depend as much on *where* you sample as on how many observations you collect.
Focus Step 3 on estimation, not thresholds. Treat Step 3 as "playing through" the planned study many times on the computer. Fit the planned analysis model and summarise what you typically get in terms of parameter estimates and uncertainty—especially uncertainty-interval widths (precision). Use bias and sensitivity to realistic departures from assumptions (e.g. increased noise, mild misspecification, measurement error) as additional diagnostics. This does not guarantee correct inference; it makes explicit what follows from the assumptions you are willing to make, and how sensitive results are to them.
Decide design targets in terms of precision and relevance. If the scientific or management claim requires an effect to be estimated within $\pm k$ units on the chosen scale, plan for an interval width of about $2k$ for the relevant parameter. Use simulation outputs (design–precision curves) to choose n, predictor range, and measurement effort that meet this target under plausible assumptions.
Report what you simulated and why. To make the workflow useful (and teachable), report the generative model, key assumptions, the range of parameters explored, the performance criterion (e.g. interval width), and the code required to reproduce results.

Checklist: reportable outputs of the simulation loop

- **Verbal hypothesis → model:** equations for $f(x; \theta)$ and the stochastic component (ε), including the response scale.
- **Assumptions:** plausible parameter values (and ranges) with justification (literature, pilot data, expert judgement).
- **Stochastic structure:** what ε represents (process variation, measurement error, both), and whether hierarchy/autocorrelation is included.
- **Design explored:** n, treatment levels, predictor range, replication structure, measurement effort; what was varied and why.
- **Performance target:** interval width (precision) for key parameters; optionally bias and sensitivity to departures from assumptions.
- **Summary outputs:** design–precision curve(s) (e.g. interval width vs n, vs predictor range, vs noise).
- **Reproducibility:** code (and seed) to reproduce simulations and figures (e.g. in Supplement), plus software versions if relevant.

5.2 Common extensions

The simulation loop is not restricted to linear regression. Common extensions follow the same three steps but use a different generative model and analysis model:

Two-group designs. A two-level treatment factor is a special case: simulate group means and variability, then examine the precision of the treatment contrast (difference in means) as a function of sample size and measurement error.

Generalised responses. For counts, proportions, presence/absence, and rates, specify an appropriate observation model (e.g. Poisson/negative binomial, binomial) and simulate on the natural scale. Precision targets may be set on the link scale or transformed back to the response scale.

Hierarchical and repeated-measures designs. Simulate nested structure (e.g. plots within sites, repeated measurements through time) and fit models with matching random effects or correlation structure. Examine how effective sample size and precision depend on the number of groups and replicates per group.

Non-linear and mechanistic models. When responses saturate, show thresholds, or depend on interacting processes, simulate under plausible parameter ranges and ensure the design includes predictor values that make key parameters identifiable (as illustrated in the non-linear example above).

Model checking after fitting. Simulation can also be used after model fitting (e.g. posterior predictive checks). In this paper we emphasise simulation earlier, when it can still inform study design and the formulation of quantitative hypotheses.

6 Outlook: why the simulation loop matters (especially with AI)

This paper offers a compact way to make simulation part of routine ecological inference: turn a verbal hypothesis into an explicit generative model, simulate what it implies, and iterate design and modelling choices based on the estimates you would obtain and their uncertainty, rather than on whether a test crosses an arbitrary threshold.

A practical motivation is that different analysis choices sometimes lead to similar conclusions, but sometimes they do not. When results are fragile to reasonable alternatives (e.g. link functions, distributional assumptions, treatment of non-independence, or measurement error), it becomes hard to know what to believe. The simulation loop provides a constructive way to examine this: simulate under plausible data-generating processes, fit candidate analyses, and quantify how estimates and uncertainty change across modelling choices. In that sense, simulation is not only a design tool but also a diagnostic for robustness and identifiability (see also simulation-based approaches to understanding and validating complex models in ecology; DiRenzo *et al.* 2023).

More broadly, choosing an analysis and deciding what counts as evidence is unavoidably epistemic: it depends on background assumptions about how data are generated, what uncertainty means, and what kinds of claims are warranted in the face of error and limited information. These questions are actively studied in philosophy of science and the foundations of statistics, but they often remain implicit in everyday practice (Romeijn, 2014; Gelman & Shalizi, 2013). Making the generative model explicit turns many hidden choices into inspectable assumptions. It also clarifies that (i) precise estimates can still be wrong when key mechanisms or error processes are misspecified (Box, 1976), and (ii) methodological choices can legitimately reflect "inductive risk"—the consequences of different kinds of error—which motivates being explicit about what uncertainty is acceptable for a given claim (Douglas, 2000; Steel, 2010). In ecology and evolution, this perspective aligns with growing calls to improve uncertainty consideration and propagation beyond narrow reliance on single metrics, and to adopt clearer standards for reporting uncertainty and the consequences of model structure (Simmonds *et al.*, 2024; Lele, 2020).

These benefits become increasingly important as ecology and evolution incorporate larger datasets, more complex models, and AI-assisted pattern discovery. Modern tools can detect subtle regularities in high-dimensional data, but they do not remove the need to decide what to believe and what uncertainty is defensible. Simulation-based thinking provides a practical way to expose and communicate assumptions, to quantify the consequences of modelling choices, and to distinguish predictive success from mechanistic understanding. In this sense, the simulation loop is both a design tool and a reasoning tool: it helps researchers and students move from compelling stories toward quantitative expectations, transparent uncertainty, and conclusions that are proportional to what the data can support.

At the same time, simulation does not guarantee correct inference: its value is conditional on the assumptions explored. It cannot by itself resolve confounding in observational studies or substitute for careful causal design (Pearl, 2019). But it can make these limitations explicit—for example, by showing that even very large datasets can yield narrow uncertainty intervals for the *wrong* quantity if key mechanisms or measurement processes are misspecified. This matters especially in AI-enabled workflows, where model complexity and opacity can make it difficult to understand how conclusions were produced, and where standards for uncertainty quantification and error propagation remain uneven across tools and subfields (Pichler & Hartig, 2023; Lucas, 2020). By encouraging explicit generative assumptions and transparent reporting of their consequences, the simulation loop offers a practical step toward more defensible inference in a future where pattern discovery will be easy, but deciding what to believe may be harder than ever.

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Box 1: Common pitfalls along the traditional scientific workflow

This box summarises common pitfalls in roughly the order they can arise along the traditional workflow shown in Figure 1. The proposed simulation loop provides a tool to anticipate and prevent many of these pitfalls.

Identifying research question

(1) Confirmation bias and motivated reasoning. Interpreting evidence in ways that align with prior beliefs or preferred narratives. Refs: Nickerson 1998; Kunda 1990; Tversky & Kahneman 1974.

(2) Weakly specified hypotheses and undefined alternatives. Hypotheses may be vague, lack a clear direction or quantitative prediction, and the implied alternative to the null is often left undefined (e.g. "there is a difference"), which makes interpretation and design targets ambiguous. Refs: Rajtmajer *et al.* 2022; Gigerenzer 2004; Perezgonzalez 2015.

(3) Confounding and causal overinterpretation of associations. A statistical association does not, by itself, identify a causal effect. Apparent treatment effects can arise because groups differ in other drivers (confounding), because sampling is selective, or because key variables were not measured. As a result, strong narratives can be attached to patterns that would also be expected under plausible alternative explanations. Refs: Pearl 2019; Rajtmajer *et al.* 2022.

Study design

(4) Insufficient sample sizes. Small samples frequently yield imprecise estimates and unstable results, and can exaggerate effect sizes by chance. Refs: Halsey *et al.* 2015; Ioannidis 2005; Baker 2016; Camerer *et al.* 2018.

(5) Planning for "power to detect" rather than precision/relevance. Sample size decisions that target a significance threshold rather than a decision-relevant interval width or estimation goal. Refs: Goodman & Berlin 1994; Rothman & Greenland 2018; Cumming 2013.

(6) Measurement error, imperfect detection, and misclassification. Noise and bias introduced by measurement instruments, proxies, observer effects, or detection limits can attenuate effects, inflate uncertainty, or create spurious patterns if not anticipated and modeled. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

(7) Non-independence and pseudoreplication. Spatial/temporal autocorrelation, clustering, and repeated measures can reduce the effective sample size and bias uncertainty if treated as independent. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

Analysis

(8) Overreliance on p-values and "statistical significance". Dichotomous interpretations substitute for effect sizes, uncertainty, and scientific relevance. Refs: Amrhein *et al.* 2017, 2019; Wasserstein *et al.* 2019.

(9) Multiple analysis choices: multiple testing, HARKing, and p-hacking. Many defensible analysis choices and multiple comparisons can yield different outcomes; hypotheses may be adjusted after seeing results (HARKing), and selective exploration can create apparently strong results without deliberate misconduct. Refs: Gelman & Loken 2013; Stefan & Schönbrodt 2023; Kerr 1998.

(10) Mismatch between model and data-generating process. Inference can be distorted when key features of the data are ignored or misrepresented (e.g. non-linear relationships fit as linear, wrong distribution or link, omitted random effects, ignored autocorrelation, inappropriate pooling, or unmodelled measurement error). Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

(11) Mathematical artefacts and scale/transform issues. Patterns can arise from statistical artefacts (e.g. regression to the mean) or inappropriate transformations; these can mimic biological effects. Refs: Barnett *et al.* 2005; Wolkovich *et al.* 2021.

Interpretation & Communication

(12) Overconfident "yes/no" summaries. Binary conclusions can sound decisive even when intervals are wide or model-dependent. Refs: Muff *et al.* 2022; Amrhein *et al.* 2019.

(13) Misreading "non-significant" as "no effect". Non-significant results are often compatible with effects that are biologically meaningful but imprecisely estimated. Refs: Goodman & Berlin 1994; Rothman & Greenland 2018; Singal *et al.* 2017.

(14) Selective reporting and publication bias (significance filtering). Significant/tidy results are more likely to be emphasized and published, distorting the evidence base. Refs: Yang *et al.* 2023; van Zwet & Cator 2021.

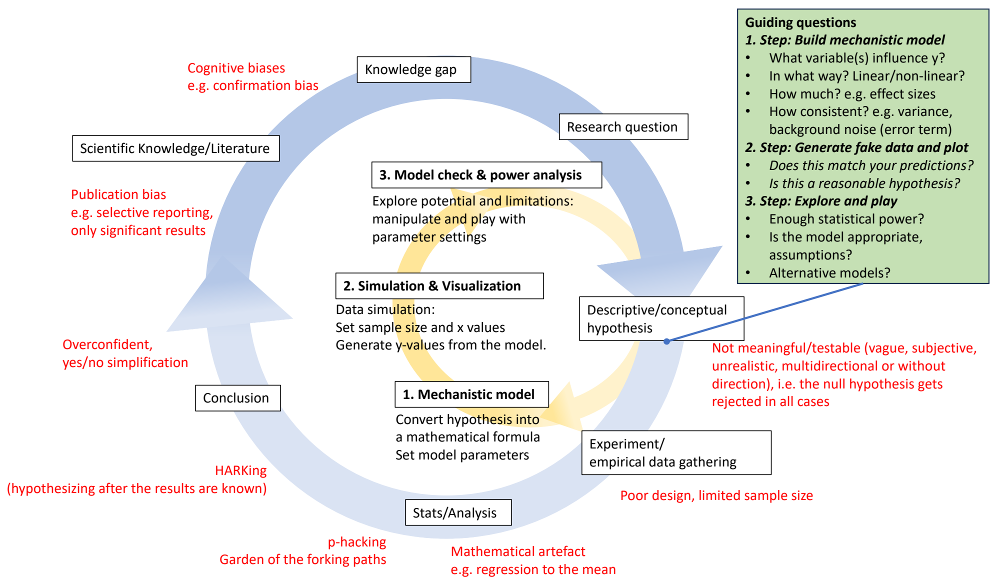

Figure 1: Schematic overview of the current scientific workflow (blue circle) with its related problems (red text) and how some of them can be anticipated with the help of data simulation (three-step-loop in yellow). The critical step is the translation of a verbal hypothesis into a mechanistic model expressed in mathematical terms. We believe that this 'simulation loop' fosters a better understanding of underlying processes founded in theory, clear predictions that can be evaluated and overall better science. The questions summarized in the green box should be addressed in the simulation loop before continuing on the blue circle, e.g. prior to the start of real data gathering/exploration.